Dark Matter Behaves Like Matter: A Halo Rule Set by Starlight Predicts Dwarf Galaxies Better Than MOND and Two Modified Gravity Theories

Gary Abraham Bernstein

ORCID: https://orcid.org/0009-0009-1761-2867

Abstract

The missing mass of galaxies behaves like matter because its amount follows the stars: a halo rule with three shared numbers, driven by stellar mass and size, predicts galaxies it has not seen, while laws that change gravity fail when carried between populations. Dark matter and modified Newtonian dynamics (MOND) both fit spirals; this paper tests their predictions on 286 systems. The halo rule predicts unseen spirals as well as MOND's acceleration relation (0.1533 against 0.1531 dex) and, unchanged, 44 dwarf spheroidals 28 percent better (0.479 against 0.669 dex) and 25 more outside the original sample 45 percent better, beating even MOND refitted on dwarfs (0.548 dex); dwarf errors remain about a factor of 3. In one likelihood over 85 dwarfs with upper limits, it beats MOND's relation by 36 log-likelihood units; its forecasts for 264 unmeasured dwarfs are hashed. Frozen on rotation curves, it predicts the amplitude of Kilo-Degree Survey lensing of 259,383 isolated galaxies inside 0.3 Mpc within 0.05 dex per stellar-mass bin, but not its radial profile or early-late split. MOG and Mashhoon's nonlocal gravity, with published constants, act as fixed copies of visible mass, strengthening gravity at a half-mass radius at most 19 and 23 times; 33 and 31 of the 44 dwarfs need more. In clusters, the published Moster halo relation with its scatter, fitted to no cluster, beats MOND's relation (0.288 against 0.403 dex) without recovering cluster masses. MOND's hot-gas cluster repair, carried to

gas-rich spirals, raises speed errors from 12 to 100 percent.

Keywords: dark matter; modified gravity; MOND; radial acceleration relation; dwarf spheroidal galaxies; galaxy clusters; nonlocal gravity; MOG; stellar-to-halo mass relation; weak lensing; cross-validation; blind test; forecasts

1. The question and the claim

The missing mass of galaxies behaves like matter because its amount follows a galaxy's stars. A halo predicted from stellar mass and size forecasts galaxies it has not seen, while the laws that change gravity tested here fail when carried from one kind of system to the next.

Galaxies rotate too fast for their visible matter. Two kinds of answer compete. One adds matter: each galaxy sits in a halo of dark matter, and the halo's mass and concentration follow from how structure formed (Navarro, Frenk and White, 1997). The other changes the law: gravity strengthens below an acceleration scale a0, as in modified Newtonian dynamics, MOND, whose radial acceleration relation ties the observed acceleration in disk galaxies to the acceleration of their ordinary matter alone (McGaugh, Lelli and Schombert, 2016; Lelli et al., 2017).

On disk galaxies both answers fit well, which is why the question has stayed open. The radial acceleration relation needs one constant, and halo fits need two per galaxy. A fair test needs what a fit to one population cannot give: predictions for systems the model has not seen, from measured inputs only, carried across populations whose sizes, gas content and histories differ.

This paper makes that test across 286 systems, at radii from 27 parsecs in dwarf galaxies to 1,000 kiloparsecs in clusters, and finds that the missing mass behaves like matter in an amount that follows the stars. A halo rule with three shared coefficients, driven by stellar mass and size, predicts unseen disks as well as MOND's relation and transfers to dwarf galaxies far better. Theories that make

extra pull from visible matter by a fixed recipe, the form two published modified gravity theories take in the weak field, cannot supply what most dwarfs need, because a fixed recipe can raise gravity only by a capped factor. A single acceleration scale, or a single rule for the extra mass, fitted where it works, fails when carried to the next population. A published halo relation set by starlight, carried unchanged to cluster scales, predicts accelerations there with 29 to 54 percent less error than MOND's relation, the smaller gain when the halo relation's scatter is included, though not the clusters' masses.

This paper's main advance is a halo rule that predicts galaxies it never saw: set by a galaxy's stellar mass and size with three shared numbers, it beats MOND on dwarf galaxies. Comparisons usually fit the galaxies they score; here every rule is fixed on some systems and scored on others, and carried from spirals to dwarfs, early-type galaxies and clusters. Two more results follow that a fit cannot give: a bound explains why modified gravity that copies visible mass by a fixed recipe with constant parameters, MOG and nonlocal gravity included, falls short on most dwarfs; and the 2025 repair of MOND in clusters cannot be extended from hot gas to the cold gas of spirals. A last result is a warning: cluster benchmarks scored at R_{500} are circular, since that radius is defined by the mass they test. The paper then scores dwarfs no rule had seen, and all 85 dwarfs together with their upper limits, under protocols hashed before scoring, and fixes forecasts for 264 dwarfs whose motions have not been measured, so that later measurements can confirm or refute the rule. A fifth protocol carries the published Moster relation, unchanged, to cluster scales as an exploratory extension, and a sixth and a seventh audit that test. An eighth and a ninth check the rule's whole halos, and the lensing profiles it predicts, against galaxy-galaxy lensing.

2. Data and the rules of the test

2.1 Five populations

The main sample holds 279 systems in five populations, and seven more clusters serve as a separate stress test.

Rotating disk galaxies come from SPARC (Lelli, McGaugh and Schombert, 2016): 175 galaxies and 3,391 radial measurements. Selecting quality flag below 3, inclination at least 30 degrees, positive radius and speed, relative speed error at most 0.1 and positive inward baryonic acceleration leaves 149 galaxies and 2,700 measurements. Disk and bulge mass-to-light ratios stay fixed at 0.5 and 0.7 in solar units at 3.6 micrometers, with catalog distances and inclinations. The baryonic speed is V_bar^2 = V_gas|V_gas| + 0.5 V_disk|V_disk| + 0.7 V_bulge|V_bulge|, and the accelerations are g_bar = V_bar^2/r and g_obs = V_obs^2/r.

Dwarf spheroidal galaxies come from McConnachie's catalog of nearby galaxies in its January 2021 update (McConnachie, 2012): 25 classical and 19 ultrafaint systems, divided at a V-band luminosity of 10^5 suns. Each gives one point at the half-light radius in three dimensions, r_1/2 = (4/3) R_e, with g_obs = 3σ^2/r_1/2 from the line-of-sight dispersion σ (Wolf et al., 2010) and g_bar = G(M_star/2)/r_1/2^2, the stellar mass set at twice the V-band luminosity. Mass-to-light ratios of one and three are kept as sensitivities. Unresolved dispersions, flagged tidal systems and measured gas above a tenth of the stellar mass are excluded.

Sixteen early-type galaxies give one inner Jeans estimate and one outer neutral-hydrogen estimate each (Lelli et al., 2017), with a photometric mass-to-light ratio of 0.8 at 3.6 micrometers. Fifty brightest cluster galaxies from MaNGA give isotropic Jeans estimates (Tian et al., 2024). Twenty CLASH clusters give 84 lensing estimates, at their centers and at 100, 200, 400 and 600 kiloparsecs (Tian et al., 2020). Seven X-COP clusters give 21 hydrostatic

estimates at fixed radii of 500, 750 and 1,000 kiloparsecs (Ettori et al., 2019; Eckert et al., 2022).

The accelerations for dwarfs, early-type galaxies, cluster centers and clusters are inferences, each carrying its source's dynamical or lensing model. They are treated as data with those assumptions attached, and no score below pools populations into one ranking.

2.2 Held out by system

Every score is a prediction for systems left out of the fit. Folds hold out whole galaxies or whole clusters, all radii of a system stay in one fold, and objects that may share a host stay together. Population tests go further: a rule fitted on one population is frozen and applied to another, with no target-population calibration. Coefficients are refitted inside each training split, each galaxy weighs equally in fits and scores, and the reported score averages three complete assignments of galaxies to folds.

Scores are root-mean-square errors in log10 acceleration, in dex. An error of 0.15 dex is a factor of 1.41; 0.48 dex is a factor of 3. Speed errors take the square root of the acceleration ratio, so a 0.15-dex acceleration error is 0.075 dex in speed, about 16 to 19 percent. The median speed errors quoted below are a different statistic, the median absolute fractional error.

The data had been inspected before these comparisons, and each protocol was fixed before its own run. The results on the main sample are exploratory comparisons of declared rules, not a preregistered discovery. The out-of-sample tests, the combined score, the two lensing checks, the cluster-scale test and its two audits (Sections 3.3 to 3.6, 3.8 and 5.3) ran under written protocols, each hashed before its own scoring, on data inspected in earlier runs. A hash fixes a file's content, not when it was written; the times given are the author's records.

3. Halos set by starlight

3.1 The rule

Each galaxy gets a finite NFW halo, density ρ_s/[z(1+z)^2] with z = r/r_s, truncated at the radius r_200 enclosing 200 times the critical density. Its concentration c_200 = r_200/r_s follows the published Planck relation of Dutton and Macciò (2014),

c_200 = 10^0.905 (M_ref / 10^12 h^-1 M_sun)^-0.101, h = 0.671,

so only the halo's reference mass M_ref needs predicting. The rule predicts it from the galaxy's stellar mass M_star = 0.5 L_3.6 and its effective radius R_e:

M_ref / M_sun = M_p/(f_b M_sun) + 10^A (M_star/10^10 M_sun)^B (R_e/3 kpc)^C,

where M_p = M_star + 1.33 M_HI is the galaxy's baryonic mass and f_b = 0.0490/0.3175 is the cosmic baryon fraction. The first term keeps the halo at least as massive as the baryons allow. The halo's dark mass is (1 - f_b) M_ref, and its pull adds to the measured baryonic pull. Fitted to all 149 disks, A = 11.748, B = 0.818 and C = -0.712; every prediction below uses coefficients fitted without the system predicted.

No rotation measurement of a predicted galaxy enters its prediction: stellar luminosity, gas mass and size set the halo, and the halo and the baryons set the curve. Stellar-to-halo relations have long precedent (Moster, Naab and White, 2013), and baryon-halo scaling has been tied to the radial acceleration relation before (Navarro et al., 2017; Li et al., 2019). What is tested here is whether such a rule, with three coefficients, carries from disks to systems it never saw. B and C nearly cancel (B + C = 0.11), so the rule depends mainly on how compact the stars are, M_star/R_e, a measure of the depth of the stellar potential. One reading is that a deeper halo pulls cooling gas into a more compact galaxy; this paper

does not test it.

3.2 Disks: as good as MOND's relation

Table 1. Shared predictors fitted on the disks: error on unseen disks and, frozen, on the 44 dwarfs, in dex.

Shared predictor	Coefficients	Unseen disks, dex	Frozen dwarfs, dex
Radial acceleration relation, refitted a0	1	0.15309	0.66937
Halo set by baryonic mass	3	0.15323	0.66757
Published Moster relation, no fitting	0	0.17111	0.55591
Halo set by stellar mass and size	3	0.15329	0.47914
Curved stellar relation	5	0.15380	0.47874

Fitted on all the disks with each galaxy weighted equally, the relation takes a0 = 1.04 × 10^-10 m/s^2 and the baryonic rule A = 11.467, B = 0.994 and C = -0.635. On unseen disks the stellar rule matches the radial acceleration relation to 0.0002 dex, and its median speed error is 9.09 percent. Two predictors built on opposite assumptions describe disk rotation equally well from the same inputs, which is why disks alone cannot settle the question. A curved stellar relation with five coefficients adds nothing.

3.3 Dwarfs: 28 percent better

Frozen on the disks and applied to the 44 dwarf spheroidals, using only their stellar mass and size, the stellar rule cuts the error from 0.669 dex for the radial acceleration relation to 0.479 dex, 28 percent lower. A paired bootstrap over whole dwarfs puts the gain over the baryonic-mass halo at -0.188 dex, with a 95 percent interval of -0.281 to -0.083. Thirty of the 44 dwarfs improve. The relation does no better when its own constant is refitted on the dwarfs and scored on dwarfs held out: 0.548 dex (Section 4.3). The stellar rule was proposed after the

baryonic rule's dwarf transfer had been seen; its coefficients come from the disks alone.

The transfer holds on a second compilation. On a fixed version of the Local Volume Database (Pace, 2025), which gives each measurement with its source, 42 dwarf spheroidals with measured dispersions, 33 of them among the 44, give the same ranking. With the same circularized radii and frozen rules, the stellar rule scores 0.389 dex against 0.611 for the radial acceleration relation and 0.612 for the baryonic-mass halo, 36 percent lower than the relation; with major-axis radii, 0.367 against 0.613 and 0.616.

A test on dwarfs outside the original sample answers the order of choices. Under a written protocol, with the rules frozen on the disks, the 25 dwarf spheroidals in the same database that are not among the 44 and pass the same selection were scored; 17 of them were scored blind, under a protocol fixed and hashed before any was seen. The stellar rule scores 0.441 dex against 0.808 for the radial acceleration relation and 0.802 for the baryonic-mass halo, 45 percent lower; the paired bootstrap interval of its gain over the relation, -0.449 to -0.272 dex, excludes zero, and 21 of the 25 dwarfs improve. On the 17 scored blind the figures are 0.405 against 0.776. The ranking holds for major-axis radii and mass-to-light ratios of 1 and 3. The published Moster relation, a second halo set by starlight and fitted to no rotation data, does slightly better on these dwarfs, 0.403 dex (0.397 on the 17 scored blind), and worse on the original 44 and the second compilation, 0.556 and 0.463 dex. Both starlight rules beat MOND's relation on all three dwarf sets.

A learning curve shows the transfer is learned from the disks, not tuned to any of them. Holding out a random 5 to 90 percent of the disks, 50 times at each fraction, and refitting every rule on the rest, the stellar rule beats the radial acceleration relation on both dwarf sets in every draw from 30 training disks up, and in at least

39 of 50 draws even from 15. As training grows, its dwarf errors settle, from 0.500 to 0.478 dex on the original 44 and near 0.44 on the 25 out of sample, their spread across draws shrinks five- to sevenfold, and its three coefficients converge; on the held-out disks the three rules stay tied.

MOND's defenders answer that a satellite's own gravity is weakened by its host's field, the external field effect, which Chae et al. (2021) report in the rotation curves of disk galaxies. Applied as McGaugh and Milgrom (2013) apply it to Andromeda's dwarfs, with the host's field V^2/D, V = 230 km/s for Andromeda and 200 km/s for the Milky Way, it lowers MOND's predictions for dwarfs that already move faster than MOND predicts: its error rises to 0.96 dex on the original 44 and 1.20 on the 25 out of sample, and stays between 0.91 and 1.00 on the 44 for Milky Way speeds of 175 to 220 km/s. On the 25 brighter dwarfs of the original sample it matches the stellar rule, 0.41 against 0.40 dex; on the 19 ultrafaint ones it misses by 1.39 dex. MOND's answer for those, that they are out of equilibrium, needs evidence that one point per dwarf cannot give.

The halo has an environment too. In simulations, a subhalo's concentration rises toward its host's center by a factor 1 - 0.54 log10(R/R_200) (Moliné et al., 2017). The factor describes concentration at a satellite's present distance, not stripping along its orbit. Applied to the stellar rule as a radial factor on its baseline concentration, with that published coefficient and host masses of 1.0 x 10^12 and 1.5 x 10^12 suns, it lowers the error to 0.457 dex on the original 44 and 0.383 on the 25 out of sample, below the Moster relation on both (0.556 and 0.403), mainly by lifting the ultrafaint dwarfs nearest their hosts. It was tried after the residuals were seen to grow toward the host, so it was kept out of the rule as a lead for the next blind test, which rejected it (Section 3.4). In an exploratory check with measured pericenters (Pace, Erkal and Li, 2022), the stellar rule's residuals for 31 Milky Way satellites show no trend with pericenter or present distance at fixed

luminosity (rank partial correlations 0.05 and -0.11), while at fixed pericenter fainter dwarfs move faster than predicted (-0.41, p = 0.02). The faint dwarfs are the nearby ones (rank correlation of distance with luminosity 0.50), so the trend toward the host is a trend with luminosity. Binary stars could be part of it: in the same multi-epoch data, modeling binaries and instrumental offsets lowers Reticulum II's dispersion from 3.5 to 2.8 km/s (Minor et al., 2019), and the two binaries found in Boötes II raise its dispersion by about 25 percent, to 3.6 from 2.9 km/s (Bruce et al., 2023).

About half of what remains is measurement. The quoted errors of the dispersions and radii give 0.30 dex on the original 44 and 0.34 dex on the 25; a likelihood that separates them leaves an intrinsic scatter of 0.32 and 0.33 dex, about a factor of 2, for differences between halos and for orbits, tides and binary stars, which a rule given only stellar mass and size cannot see.

The one change between the two halo rules is which mass sets the halo: stars or all baryons. Disks cannot tell them apart. Dwarfs, which have stars and almost no gas, can. The halo follows the stars.

The gain is not uniform. Ultrafaint dwarfs improve from 0.934 to 0.571 dex, and classical dwarfs worsen from 0.348 to 0.395. Satellites of the Milky Way improve, and those of Andromeda worsen. Three of four tails of the disk sample, cut by predictor values, worsen slightly. The remaining dwarf errors, a factor of about 3, are large, and satellite evolution, tides and binary stars are not modeled. The result is a better transfer, not a solved dwarf problem.

3.4 Dwarfs no rule had scored

The database holds two kinds of dwarfs that no earlier test scored. Some lie beyond the Milky Way and Andromeda systems, and some have only an upper limit on their dispersion. Dropping the limits drops the slowest dwarfs, which

favors any rule that predicts fast motion. A third protocol, written and hashed before any prediction for these dwarfs was computed, scored both kinds with every rule frozen (Table 2).

Table 2. Dwarfs no rule had scored, every rule frozen: error on the five with measured dispersions, in dex, and upper limits exceeded, of ten.

Model	Five measured, dex	Limits exceeded
Halo set by stellar mass and size	0.171	2
Stellar rule with subhalo factor	0.171	3
Radial acceleration relation, disk a0	0.202	0
Radial acceleration relation, published a0	0.185	0
MOND with external field	0.208 (four dwarfs)	0
Halo set by baryonic mass	0.225	0
Published Moster relation	0.235	4

Five dwarfs have measured dispersions: Cetus, Phoenix, Tucana and UGC 4879, which have no host galaxy, and KDG 64 in the M81 group. The first four are far from any large galaxy, so they feel almost no tides or external field, and neither of MOND's answers for satellites applies to them. The stellar rule has the lowest error of seven models, 0.171 dex against 0.202 for MOND's relation, 0.208 for MOND with the external field on the four it covers, and 0.235 for the Moster relation. Five dwarfs cannot make that significant: the bootstrap interval of the gain over MOND's relation runs from -0.170 to +0.144 dex, so the protocol's first test is not met. The ranking is also fragile. With major-axis radii, MOND's relation with its published a0 scores lower than the stellar rule, 0.186 against 0.201 dex, and at a mass-to-light ratio of 3 every version of MOND does.

Ten satellites have only upper limits, at 95 percent confidence in the database. The stellar rule's predicted dispersion lies below eight of them. It exceeds the limit of Grus II, 3.3 against 2.0 km/s (Simon et al., 2020), and of Andromeda XII, 5.2

against 4.0 km/s (Collins et al., 2013); only Grus II lies beyond the rule's own scatter, a factor of 1.46 in dispersion. MOND meets all ten, so the protocol's second test is not met either. MOND passes because it predicts slow motion for every faint dwarf. The same predictions miss the faint dwarfs with measured dispersions in Section 3.3, 19 in the original sample and 17 of the 25 outside it, by 1.39 and 1.35 dex in root-mean-square, factors above 20 in acceleration. The Moster relation, which predicts faster motion than the stellar rule, exceeds four limits. These counts hold for major-axis radii, mass-to-light ratios of 1 and 3, and Milky Way speeds of 175 and 220 km/s.

Upper limits can catch only a rule that predicts too much motion. Measured dispersions catch errors both ways, and on all four dwarf sets with measured dispersions the stellar rule is ahead of MOND's relation. Section 3.5 weighs both kinds in one score. The subhalo factor of Section 3.3 raises the predicted dispersions of satellites close to their hosts, where several limits are lowest, and exceeds three limits, one more than the rule without it. The lead fails its test and stays out of the rule.

3.5 All 85 dwarfs in one score

Measured dispersions and upper limits can be weighed together. A fourth protocol, hashed before any likelihood was computed, puts all 85 dwarfs scored so far into one likelihood, every value taken from the database: 75 measured dispersions count as values with their errors, and 10 upper limits count as ceilings. Each model keeps its prediction as the center, and its one free number is its scatter (Table 3).

Table 3. All 85 dwarfs in one likelihood, 75 measured and 10 upper limits, every rule frozen. Each model's only free number is its scatter, in dex. The last column is how far its log-likelihood falls below the stellar rule's.

Model	Scatter, dex	Log-likelihood	Behind stellar rule
Halo set by stellar mass and size	0.30	-39.3	0
Stellar rule with subhalo factor	0.29	-36.9	-2.4
Published Moster relation	0.40	-54.4	15.1
Radial acceleration relation, published a0	0.56	-72.8	33.5
Radial acceleration relation, disk a0	0.58	-75.4	36.2
Halo set by baryonic mass	0.58	-75.7	36.4
MOND with external field	0.93	-106.0	66.7

Among the baseline frozen rules, the stellar rule fits best, with a scatter of 0.30 dex; the exploratory subhalo variant, discussed below, scores slightly higher. It beats MOND's relation by 36 units of log-likelihood, with a bootstrap interval of 24 to 48 units, which meets the protocol's test. It beats MOND with the external field by 67 units. The limits cost the stellar rule 4 units more than MOND's relation, and the measured dwarfs give it 40 units more. On average the stellar rule sits 0.02 dex from the data, while MOND's relation sits 0.43 dex too low. Even when MOND's relation may shift its whole prediction to fit the dwarfs, the unshifted stellar rule stays ahead by 9 units.

The gain over MOND's relation stays between 27 and 48 units for mass-to-light ratios of 1 and 3, for major-axis radii, without measurement errors, and with every measured dispersion below 3 km/s dropped. The last checks binary stars, which inflate the smallest dispersions most; modeling the binaries themselves needs repeated spectra of each star and is left for later work. The Moster relation trails the stellar rule by 15 units (3 to 25). The subhalo factor scores 2.4 units better

than the rule without it, inside its bootstrap interval of -8.6 to +3.3, so its net effect is not established and it stays out of the rule.

Two checks added after the run score each dwarf held out in turn. With each model's scatter fitted to the other 84 dwarfs, the stellar rule leads MOND's relation by 35.7 units, with a bootstrap interval over dwarfs of 21.8 to 47.8, and 70 of the 85 dwarfs favor it. When both may also shift their predictions, again fitted without the dwarf scored, the lead is 8.8 units (1.4 to 15.4). With that shift the baryonic-mass halo comes within 4.2 units (-4.4 to 11.4): its loss on dwarfs is mostly one uniform shortfall of 0.45 dex, and setting that level right from the disks alone is what the stellar rule adds. These scores share the likelihood's simplifications: an upper limit counts as an exact ceiling, the errors that size and distance share between predicted and measured accelerations are left out, and the bootstrap treats satellites of one host as independent.

3.6 Forecasts for dwarfs not yet measured

The same protocol fixed the dispersion each rule forecasts for the 264 gas-poor dwarfs in the database with no measured dispersion. The file was hashed on 31 October 2026, and its SHA-256 begins bbafc29918f89053. Table 4 lists the nine in the Local Group, with 68 percent ranges from the scatter of Section 3.5. For these, the stellar rule forecasts dispersions a median 1.5 times MOND's with the external field. The protocol calls a forecast decisive when the two differ by more than a factor of 2, which among the nine holds for Carina IV alone, the faintest: 2.1 km/s from the stellar rule against 0.84 from MOND. The other 255 lie farther out, most in groups of galaxies up to 13 megaparsecs away, where the two forecasts differ by a median factor of 1.3. The ranges are a factor of 1.42 either way for the stellar rule and 2.9 for MOND, whose scatter must be wide to cover its misses. Measured together, the nine separate the two forecasts by 1.4 standard

deviations, so the set weighs more than any single dwarf. Measured dispersions near MOND's for the nine would count against the rule, and near the rule's, against MOND.

Table 4. Forecast line-of-sight dispersions, km/s, with 68 percent ranges, for the Local Group dwarfs with none measured, faintest first. MOND includes the field of the host, or for dwarfs with no host the stronger of the Milky Way's and Andromeda's fields.

Dwarf	Host	M_V	Stellar rule	MOND	Ratio
Carina IV	Milky Way	-2.8	2.11 (1.49 to 2.99)	0.84 (0.29 to 2.44)	2.50
Leo K	none	-4.73	3.09 (2.18 to 4.38)	1.80 (0.62 to 5.23)	1.71
Andromeda XXXV	M31	-5.2	3.07 (2.17 to 4.35)	2.01 (0.69 to 5.84)	1.53
Pegasus VII	M31	-5.7	4.13 (2.92 to 5.85)	2.25 (0.77 to 6.54)	1.83
Leo M	none	-5.77	3.76 (2.65 to 5.33)	2.29 (0.79 to 6.66)	1.64
Andromeda XXXVI	M31	-5.91	3.49 (2.46 to 4.94)	2.37 (0.82 to 6.89)	1.47
Pegasus V	M31	-6.3	3.91 (2.76 to 5.54)	2.59 (0.89 to 7.53)	1.51
Tucana B	none	-6.9	4.20 (2.97 to 5.95)	2.97 (1.02 to 8.64)	1.41
Pegasus W	none	-7.2	4.50 (3.18 to 6.37)	3.18 (1.09 to 9.25)	1.41

3.7 What does not help

Published feedback profiles, which change halos by the gas that star formation drives out and which fit individual galaxies well (Li et al., 2020), do not improve the shared rule: the DC14 recipe (Di Cintio et al., 2014) gives 0.189 dex on disks, a mixture with NFW chooses a 5 percent share and gains nothing, and coreNFW (Read, Agertz and Collins, 2016) with an assumed formation time gives 0.163 dex. These cored profiles are the usual answer to the cusp-core problem, and the shared rule does better without them. A blend of the radial acceleration relation and the stellar halo, weighted per training split, lowers the disk error by under 2

percent, a gain its uncertainty does not establish, and transfers to dwarfs less well than the stellar halo alone (0.505 dex). Allowing halo scatter improves a predictive log score by 0.19 units per radius, with a 95 percent interval of 0.09 to 0.30, without improving the central error; the stellar rule's gain is in the point prediction itself.

The stellar rule is calibrated on single disk galaxies and is not carried to the heavier systems. Above about the Milky Way's stellar mass, the relation between a galaxy's stars and its halo bends, as mergers add halo mass faster than stars (Moster, Naab and White, 2013), and a cluster's potential sums the halos of hundreds of member galaxies and its hot gas. Standard structure formation predicts that change; a universal force law has to hold with one constant through it, and Section 5 tests whether it does. Section 5.3 carries the published Moster relation, which includes the bend, to cluster scales.

3.8 Whole halos from lensing

Rotation curves reach a few tens of kpc, while galaxy-galaxy lensing weighs whole halos. Mandelbaum et al. (2016) measured the mean halo masses of red and blue central galaxies in the Sloan survey, in bins of stellar mass. Under an eighth protocol, hashed before any prediction, the stellar rule frozen on the disks, given the disks' own scalings of size and gas with stellar mass, predicts the halos of the blue, star-forming centrals within 0.16 dex in four of five bins from 10^10.2 to 10^11.3 suns, a median 0.05 dex high. At 10^10.56 suns it is 0.34 dex high, where the measured mass dips below that of the bin beneath, so the protocol's test, every bin within 0.2 dex or twice its error, is not met. Means over a bin exceed typical masses by roughly 0.05 to 0.1 dex, which the comparison leaves uncorrected. For red centrals half the size of disks, the rule matches the measured halos up to 10^10.9 suns but gives only about half of the measured

red-blue split there, 0.17 to 0.20 dex against 0.35 to 0.41. Above 10^11 suns the red centrals' halos grow about as the square of stellar mass and lie above the rule by 0.3 dex at 10^11.1 suns and 1.0 dex at 10^11.7: the bend of Section 3.7, reached by the centrals of groups and clusters. X-ray masses of 21 clusters show the same steepness from the other side: their central galaxies' stellar mass grows only about as M_500^0.4, with about 0.2 dex scatter (Kravtsov, Vikhlinin and Meshcheryakov, 2018). The inverted Moster relation lies 0.4 to 1.0 dex above the red centrals' mean halos over the same range.

Lensing also traces how the mass is spread. Brouwer et al. (2021) stacked the Kilo-Degree Survey (KiDS-1000) lensing signal of 259,383 isolated galaxies at redshifts 0.1 to 0.5 in 15 bins of baryonic acceleration, spanning about 30 kpc to 3 Mpc, for four bins of stellar mass and for early and late types of matched stellar mass. A ninth protocol, hashed before any measured value was read, as were the predicted profiles, carries the frozen rule into these stacks. It uses the published stellar masses, the disks' scalings of size and gas for late types, and, for early types, no gas and the median size gap between early and late types at each stellar mass in Hubble imaging (van der Wel et al., 2014). Inside 0.3 Mpc, where false isolated lenses add little, the measured signal lies 0.027 ± 0.015 dex below the prediction for the whole sample, between 0.005 and 0.048 dex below it in each of the four stellar-mass bins from 10^8.5 to 10^11 suns, and 0.00 and 0.01 dex below it for red and bulge-dominated galaxies. For blue and disk-dominated galaxies it lies 0.14 and 0.12 dex below, outside the protocol's 0.1 dex; the rule likewise exceeds the halo masses of Mandelbaum's blue centrals of 10^10.5 to 10^11 suns, by 0.13 to 0.34 dex. At redshift 0.3, near KiDS's 0.25, CFHTLenS halo-model masses of blue galaxies, with satellites modeled, instead lie within 0.07 dex of the rule at 1.7 and 4.2 × 10^10 suns (Hudson et al., 2015), so the surveys do not yet agree for these galaxies. The rule therefore puts early types

only 0.05 dex above late types of the same stellar mass, against a measured 0.16 dex by Sérsic index and 0.21 dex by color inside 0.3 Mpc: about a third and a quarter of the split, which the protocol counts as part of the first and not the second. Beyond 0.3 Mpc the whole sample's signal exceeds the single-halo model by 0.16 to 0.57 dex, where Brouwer et al. expect neighbors missed by the isolation criterion to add signal. An independent script reproduces every amplitude to 0.0001 dex.

These are amplitudes; the radial profile does not match. With the release's covariance, the whole sample's seven bins inside 0.3 Mpc give a chi-square of 48.6 against the frozen prediction and 45.1 for six degrees of freedom once the amplitude is fitted (p = 4 × 10^-8). The measured signal lies 0.19 dex below the prediction in the innermost bin and 0.16 dex above it in the bin nearest 0.3 Mpc, so the measured profile is flatter than the predicted one. Three of the four stellar-mass bins and the blue and bulge-dominated galaxies also miss the profile, at p = 10^-4 to 0.03, while the red and disk-dominated galaxies match it (chi-square 7.6 and 8.1 for six). The protocol's lensing model extends each halo beyond r_200, where the rotation-curve fits truncate it; halos truncated at r_200 move the amplitudes by 0.003 dex or less and still miss the profile (40.6 for six), as do all 14 of the protocol's sensitivity cases (39.5 to 50.1). These tests take the model as exact and use only the measurement covariance, and since the samples overlap, their p-values are not independent.

Lensing does not yet support the rule's size term at fixed stellar mass and type. In KiDS near 3 × 10^10 suns, halo mass follows size and Sérsic index more closely than color, with a suggestion that compact early types sit in heavier halos while late types share about one halo mass (Taylor et al., 2020). In CFHTLenS, at fixed stellar mass and color, halo mass grows with size, as size to the power 0.28 ± 0.18 for blue galaxies and 0.53 ± 0.15 for red ones, and 0.22 ± 0.42 for blue

galaxies near 10^10.6 suns, with satellites kept in the samples and concentration fixed by mass (Charlton et al., 2017); the rule's halo mass instead falls as size to the power of about -0.6 there. Charlton et al. note that with concentration fixed, part of a size trend could be a change of concentration instead. A lensing measurement of halo mass against size at fixed stellar mass and type, for central galaxies and with the rule's prediction fixed first, would decide it.

4. A mass budget no fixed kernel can meet

4.1 The bound

In the weak-field limit, nonlocal gravity (Rahvar and Mashhoon, 2014) and MOG with constant parameters (Moffat and Rahvar, 2013) both act as if the ordinary matter carried extra effective matter, spread from it by a fixed positive kernel q:

$\nabla^2\Phi = 4\pi G(\rho_b + \rho_{eff})$, $\rho_{eff}(x) = \int q(|x - y|)\, \rho_b(y)\, d^3y$,

with $q_{NLG}(d) = \alpha\mu(1 + \mu d)e^{(-\mu d)}/(8\pi d^2)$ and $q_{MOG}(d) = \alpha\mu^2\, e^{(-\mu d)}/(4\pi d)$. Each kernel integrates to α over all space. An isolated source of baryonic mass M_b therefore carries total effective mass αM_b, and no part of space can hold more of it than that. At the source's own half-mass radius $r_{1/2}$, inside which the baryonic mass is $M_b/2$, the enclosed effective mass is at most αM_b, so

$g/g_b = 1 + M_{eff}(< r_{1/2})/(M_b/2) \le 1 + 2\alpha$.

The bound holds for any positive spherical source, whatever the kernel's range $1/\mu$ and whatever the source's radial profile. For a source whose density falls outward, such as a Plummer sphere, no displaced ball encloses more mass than the centered one, and the ceiling tightens to $1 + \alpha$, since then $M_{eff}(< r_{1/2}) \le \alpha M_b/2$.

4.2 The dwarfs exceed it

The published fitted values (α, μ in kpc^-1) are (10.94, 0.059) for nonlocal gravity and (8.89, 0.042) for MOG, so the general ceilings are 22.9 and 18.8, and for a Plummer sphere 11.9 and 9.9. The dwarf estimates sit at (4/3)R_e, where a Plummer source encloses 0.512 of its mass, so the general ceiling there is 1 + α/0.512: 22.4 and 18.4. At a stellar mass-to-light ratio of 2, the inferred ratio g_obs/g_bar of 31 of the 44 dwarfs exceeds the nonlocal gravity ceiling, and 33 exceed MOG's. At a mass-to-light ratio of 3, 29 exceed each. These are counts of central estimates, and the estimates carry the assumptions of dynamical equilibrium, membership, anisotropy and binary stars; but the bound itself needs no fitting, and most dwarfs sit well above it.

4.3 Integrating the kernels

The bound is a ceiling. The full prediction integrates each kernel against a Plummer sphere fixed by each dwarf's stellar mass and size, with matter outside the half-mass radius included:

Table 5. Each dwarf's stars spread as a Plummer sphere with half its mass inside r_1/2, with the relation and the two kernels integrated over it, in dex. Refitted models are scored on dwarfs held out in five folds, in the first of three assignments; the mean of the three is 0.543 for the relation and 0.722 for the kernels.

Model, mass-to-light ratio 2	Dwarfs held out, dex
Published radial acceleration relation, fixed a0	0.651
Radial acceleration relation, a0 refitted on dwarfs	0.548
Published nonlocal gravity kernel	1.862
Published MOG kernel	1.905
Nonlocal gravity, both coefficients refitted	0.729
MOG, both coefficients refitted	0.729

The relation scores 0.651 here with its published a0 and 0.669 in Section 3.3 with a0 refitted on the disks. Here the Plummer scale puts exactly half the mass inside r_1/2, where Section 4.2 takes the scale as R_e, which puts 0.512 inside; an independent integration with that scale and its own folds agrees with the table within 0.01 dex.

With their published coefficients the kernels miss by factors near 70. Refitted on the dwarfs, all 108 fits drive the range parameter μ to its upper bound, collapsing the kernel to a point and the theory to a constant multiplier of about 59; even then they do worse than the radial acceleration relation. Earlier studies found the same tension by other routes: no common MOG parameter pair for eight classical dwarfs with plausible stellar masses (Haghi and Amiri, 2016), and tensions in the nonlocal kernel's parameters in full Jeans analyses (De Martino, Della Monica and Roshan, 2025). The bound shows why. The extra pull a fixed linear kernel can give is capped by the matter it is given, and dwarfs have too little matter for their pull.

The bound covers isolated sources and constant coefficients. Running-parameter MOG, external fields, nonlinear responses and full covariant theories need their own treatment.

5. One acceleration scale does not carry across

5.1 The scale moves with the population

Fitted on the disks point by point, the radial acceleration relation takes a0 = 1.16 x 10^-10 m/s^2. Frozen at that value and carried to the other populations, its median acceleration errors are 50 percent for classical dwarfs, 86 percent for ultrafaint dwarfs, 34 percent for early-type galaxies, 48 percent for cluster centers and 69 percent for CLASH clusters. Refitted on all five populations together, a0 becomes 5.34 x 10^-10 and the disks lose their accuracy. Fitted on the X-COP

clusters alone, it becomes 7.16 x 10^-10, six times the disk value. A law with one constant that needs a different constant for each kind of system is no longer one law. That clusters need more than MOND supplies is known (Tian et al., 2020; Famaey, Pizzuti and Saltas, 2025); the numbers here put the disk-to-cluster gap on one footing with the rest.

Freer formulas do not repair it. A six-coefficient quadratic in acceleration and radius wins when all populations are in training, 0.236 dex on held-out groups, and falls to 0.488 dex when a whole population is held out. Flexibility fits the populations it has seen.

5.2 Extra mass that follows hot gas, and not cold

Famaey, Pizzuti and Saltas (2025) proposed repairing MOND in clusters with an extra density that follows the hot gas, $\rho_{extra} = \eta\, \rho_{hot}\, e^{(-r/R_c)}$. With a_0 fixed at 1.2×10^{-10} and only η and R_c fitted, the repair transfers between cluster surveys: trained on 16 CLASH clusters ($\eta = 11.8$, $R_c = 289$ kpc), it predicts the seven X-COP clusters with a median error of 7.6 percent, against 53 percent for a single refitted a_0. The reverse transfer gives 17 against 38 percent, though within CLASH alone the repair does no better than one refitted a_0.

The repair is stated for hot gas. Carrying it to gas-rich disks tests whether it can become a rule for gas as such, whatever its temperature. In 61 disks where gas supplies at least half the baryonic pull, the fit prefers η near zero, with an 11.6 percent median speed error; carrying the cluster value over raises the speed error to 99.7 percent. Forced to share η and R_c, disks and clusters both deteriorate, and the joint fit pushes R_c to its upper bound. The extra mass appears where the gas is hot, inside deep cluster potentials, and not where it is cold, which is how a component tied to structure formation behaves and how a rule attached to gas does not.

5.3 A published halo relation at cluster scale

The halo side can be carried across the same gap, as an exploratory extension that rests on the published Moster relation alone; the paper's own three-coefficient rule remains a disk-to-dwarf result. The Moster relation of Sections 3.2 to 3.5, unchanged, maps a central galaxy's stellar mass to its host halo's mass up to about 10^15 suns. A fifth protocol, hashed before any prediction was computed, applies it with nothing fitted to the 50 MaNGA brightest cluster galaxies and the 20 CLASH clusters of Section 2.1. Each brightest galaxy's published stellar mass and redshift set its halo, the concentration follows Dutton and Macciò (2014) at that redshift, and the measured baryons are added (Table 6); the CLASH stellar masses are those of Cooke et al. (2016), which Tian et al. take as total. Moster, Naab and White (2013) define halo mass at 200 times the critical density, as the concentration relation does, and give masses in suns for h = 0.704; as in Section 3.1, the halo's dark mass is (1 - f_b) times the halo mass, and its pull adds to the measured baryonic pull. Each galaxy or cluster weighs equally.

Table 6. Cluster-scale errors in dex with nothing fitted to these systems: the published Moster relation, inverted at its mean (fifth protocol) or with its scatter and the halo mass function (sixth protocol), MOND's relation and the disk-calibrated stellar rule.

Model	All 70 systems	MaNGA BCGs	CLASH centers	CLASH, 100 to 600 kpc
Published Moster relation, mean	0.185	0.182	0.154	0.204
Published Moster relation, with scatter	0.288	0.240	0.214	0.422
Radial acceleration relation, disk a0	0.403	0.338	0.479	0.550
Radial acceleration relation, published a0	0.388	0.327	0.460	0.526
Stellar rule, disk-calibrated	0.474	0.328	0.435	0.791

Inverted at its mean, the Moster relation cuts the error of MOND's relation by more than half, 0.185 against 0.403 dex, and the bootstrap interval of the gain, -0.248 to -0.188 dex, meets the protocol's test. Its median acceleration error is 26 to 33 percent in each group, against 50 to 72 percent for MOND's relation, which falls short in every group: by 0.30 dex in the MaNGA galaxies, 0.45 dex at CLASH centers and 0.54 dex at 100 to 600 kpc. The Moster relation's largest offset is at the MaNGA centers, 0.15 dex short. The disk-calibrated stellar rule fails at 100 to 600 kpc (0.79 dex), as Section 3.7 expects, since it has no bend above the Milky Way's mass. The gain holds with the CLASH stellar masses doubled, with the MaNGA masses lowered by 0.3 dex, and without redshift evolution: 0.20 to 0.24 dex against 0.40.

The relation is nearly flat at the top, so a small change in stellar mass moves the halo mass a lot. Five of the 20 CLASH galaxies map to halos above 10^16 suns, beyond any known cluster, and these five are overpredicted by 0.02 to 0.46 dex. For two of them, MACS J0429.6-0253 and MACS J0647.7+7015, the search

found no root and stopped at its upper bound, 10^16.5 suns. Without redshift evolution only one halo lies above 10^16 suns, and the error falls to 0.20 dex.

A sixth protocol, hashed before it ran, audited this test. Inverting the mean relation ignores its published scatter of 0.15 dex in stellar mass at fixed halo mass. Inverted instead with that scatter, the measurement errors and the halo mass function of Tinker et al. (2008), computed with Colossus (Diemer, 2018), the relation puts every halo's posterior median below 10^15 suns and still beats MOND's relation, 0.288 against 0.403 dex, a 29 percent cut rather than 54, with a bootstrap interval of the gain of -0.136 to -0.095 dex (Table 6). Scored only on the 50 systems whose stellar masses lie within the range the relation was fitted on, at most 10^11.7 suns, the mean inversion gives 0.187 against 0.377 dex, and with h = 0.704, as Moster's parameters assume, 0.182 against 0.403. The accelerations survive these checks; the halo masses do not.

Tian et al. computed the CLASH lensing accelerations from the spherical NFW fits of Umetsu et al. (2016), and a seventh protocol, hashed before it ran, compares the Moster masses with those published masses, M_200 of 10^14.8 to 10^15.5 suns; NFW fits to the accelerations themselves recover the published masses within 0.05 dex. The mean inversion's masses scatter by 0.73 dex about the published ones, a factor of five, where the published errors allow 0.13 dex, around a median 0.21 dex high; only 5 of the 20 lie within twice the published error. No central galaxy could do much better: with its stellar mass growing as about M_500^0.4 and scattering by about 0.2 dex (Kravtsov, Vikhlinin and Meshcheryakov, 2018), it fixes the halo mass only to about 0.5 dex, while the cluster's total stellar mass, growing as M_500^(0.6 ± 0.1) with 0.1 dex scatter, would fix it to about 0.2 dex. The probabilistic masses fall a median 0.98 dex low, and none lies within twice the error, as expected when the mass function's prior ignores how CLASH chose its clusters: 16 for X-ray temperatures above 5 keV

and smooth, symmetric X-ray images, and 4 for their high lensing magnification (Umetsu et al., 2016). The four chosen for lensing fall furthest, 1.43 dex low against 0.89 for the sixteen. With halos that light, the probabilistic predictions fall short in every group, by 0.18 to 0.40 dex. Lensing puts the mean halos of red central galaxies 0.4 to 1.0 dex below the inverted relation above 10^11.1 suns (Section 3.8), so part of the mean inversion's lead over the probabilistic one may come from that overshoot offsetting CLASH's selection of massive clusters. At these radii a halo's pull changes by only about 0.2 to 0.7 dex per dex of halo mass, and the baryonic pull not at all, so the test shows that halos set by starlight predict cluster-scale accelerations better than MOND's relation, not that the relation predicts each cluster's mass. The same published relation is within 12 percent of MOND's relation on disks and ahead of it on the dwarfs taken together, while MOND's relation needs its constant raised about six times between disks and clusters.

6. Why cluster benchmarks at overdensity radii are circular

Scoring acceleration relations in clusters at overdensity radii such as R_500 or R_200 builds in a circularity that can pass for unification. In the X-COP clusters, a shared acceleration-and-radius formula scored at R_500 shows a 5.63 percent median error on held-out clusters, and the accuracy comes from the definition of R_500. That radius encloses 500 times the critical density ρ_crit = 3H(z)^2/(8πG), so M_500 = (4π/3) 500 ρ_crit R_500^3, which fixes the target acceleration at g_500 = G M_500/R_500^2 = 250 H(z)^2 R_500. A formula in radius and redshift alone reproduces the target to 0.0001 percent, with no baryonic information. Holding out whole clusters does not break an identity built into the radius. The clusters here are therefore scored at fixed physical radii, and any cluster benchmark scored at R_500 or R_200 should be checked for the same identity.

The circularity concerns tests that predict the acceleration at a radius built from the same mass; measuring a cluster's mass at R_500 is not circular.

7. What it means

Three results point the same way. A halo whose mass follows the stars predicts disks as well as MOND's relation and dwarfs better, with no rotation data from the predicted galaxy, and a published halo relation set by starlight, with nothing refitted, predicts cluster-scale accelerations better than MOND's relation, though not the clusters' masses. A modification that turns ordinary matter into extra effective matter through a fixed kernel cannot reach what most dwarfs need. One acceleration scale, or one rule for extra mass attached to gas, fitted where it works, fails when carried to the next kind of system. The missing mass behaves like matter: it has its own amount, set by how the galaxy formed and traced by its stars, and gravity acts on it as on everything else.

Description length, counted roughly in coefficients rather than bits, gives the same ranking from another side: a description earns its weight by what it predicts per bit. A halo rule of three coefficients that carries from disks to dwarfs is shorter, for what it predicts, than a force law that needs a new constant or a new extra component for each population. The results fit a broader pattern: short laws hold with no built-in scale, as $\nabla^2\Phi = 4\pi G\rho$ does, and the variety lives in the state the laws produce over time. A force law that builds a scale into itself, through a kernel's range or an acceleration threshold, puts the complexity in the law, and here such laws failed when carried to systems of different density and history. A dark component that obeys the ordinary law gathers where baryons cool in its potential well, which would explain why a galaxy's stars and size predict its unseen mass. A search for a short fundamental rule, or for a hidden sector, can take these numbers as targets: whatever the dark component is, it must collect

around stars in the proportions found here.

The comparison has limits. It fits no raw lensing shear, cluster temperature, cosmic microwave background or large-scale structure likelihood; a complete case for either side must, and relativistic MOND theories already compute cosmological spectra (Skordis and Zlosnik, 2021). Nor does it address the S8 tension, the lower clumpiness some weak lensing surveys find than the cosmic microwave background predicts (Pantos and Perivolaropoulos, 2026). Whatever settles it must still produce halos that follow starlight as found here. MOND's external-field effect enters only through the estimate of McGaugh and Milgrom (2013), which takes the host's field as uniform; a full solution along each satellite's orbit is left for later work. Dwarf masses rest on equilibrium estimators. The halos use a median concentration relation and no formation histories. Each protocol was fixed before its run on data inspected in earlier runs. The cluster-scale test models no cluster selection and does not recover halo masses. Within those limits the three tests agree.

A fit cannot prove what dark matter is. Proving it would take detecting the dark component directly, or a rule that reproduces every population to measured precision. Ideally the rule would also predict something new that is then confirmed. Section 3.6 makes such predictions and fixes them with a hash.

The next tests follow. Spectra of the nine Local Group dwarfs of Table 4 check the forecasts directly, and tighter measurements of Grus II and Andromeda XII will show whether the rule's two misses are scatter.

Brouwer et al. (2021) find that, at the same stellar mass, early types show lensing accelerations 0.17 dex (split by Sérsic index) to 0.27 dex (by color) above those of late types, a difference of 5.7 sigma or more. Its authors note that a gravity modification independent of other galaxy properties could not explain this, and that circumgalactic gas equal to the stellar mass around the early types alone

would reduce it to about 4 sigma, since equal stellar mass need not mean equal baryonic mass. The stellar rule's size term gives a quarter to a third of the split (Section 3.8); the rest comes from massive late types whose lensing falls below the rule. Two tests follow: halo mass against size at fixed stellar mass and type, where the rule predicts lighter halos for larger disks, and a halo whose concentration, rather than mass, depends on size, fitted to rotation curves and scored on lensing; the lensing profile, flatter than predicted (Section 3.8), also calls for a freer concentration. Dwarfs with full Jeans profiles and modeled external fields should keep the stellar rule ahead, or overturn it. A fuller halo model should be scored the same way: the bend in the stellar-to-halo relation above the Milky Way's mass, size measured against its expected value at fixed stellar mass, size-dependent concentration, tidal stripping of satellites, using measured Gaia orbits where they exist, and unresolved binary stars, which inflate the dispersions of the faintest dwarfs (McConnachie and Côté, 2010). A modified gravity law that hopes to cover all scales must pass the mass budget of Section 4 and one constant across disks, dwarfs and clusters.

8. Conclusion

Across 286 systems, predicted from measured inputs and scored on systems left out, the missing mass of galaxies behaves like matter whose amount follows the stars. A three-coefficient halo rule driven by stellar mass and size matches MOND's radial acceleration relation on unseen disks and predicts dwarf galaxies 28 percent better, and 25 dwarfs outside the original sample 45 percent better. On dwarfs no rule had scored, under a hashed protocol, it has the lowest error of seven models on five with measured dispersions and stays below 8 of 10 upper limits. Scored in one likelihood, all 85 dwarfs with their limits favor it over MOND's relation by 36 units, and its forecasts for 264 dwarfs not yet measured are fixed,

with ranges, for the next spectra to check. Fixed kernels of nonlocal gravity and MOG with constant parameters cannot exceed a factor 1 + 2α at a source's half-mass radius, and most dwarfs need more. The remaining dwarf errors, about a factor of 3, half of it measurement, leave room for a fuller halo model. Frozen on rotation curves, the rule also predicts the lensing halo masses of star-forming central galaxies within 0.16 dex in four of five stellar-mass bins. Under a ninth protocol it predicts the amplitude of the stacked KiDS lensing of isolated galaxies inside 0.3 Mpc within 0.05 dex in every stellar-mass bin and for early types, but not its radial profile; massive late types fall 0.12 to 0.14 dex below it, and the stars' size gives only a quarter to a third of the split between early and late types. MOND's fitted acceleration scale grows six times from disks to clusters, and its hot-gas cluster repair cannot be carried to cold-gas disks. In an exploratory extension, the published Moster relation, unchanged, predicts cluster-scale accelerations with 29 to 54 percent less error than MOND's relation, the smaller gain when the Moster relation's scatter is included, though not each cluster's mass. In these tests, halos set by starlight carry across and the universal force laws do not. The law is the same everywhere, and the variety lives in the matter it acts on.

Data availability

The supplement contains the frozen inputs, selection and exclusion tables with provenance and hashes, all coefficients, the prediction tables for every fold, the protocols fixed before each run, independent numerical checks and the scripts that reproduce every number here. The disk fits also rebuild from the public SPARC files with independent code, and separate scripts rerun the dwarf test on a fixed version of the Local Volume Database, run the out-of-sample test, with both versions of its protocol and their hashes, the learning curve, the

external-field, subhalo and error tests, and the third protocol's test on dwarfs no rule had scored, with its forecasts for 264 dwarfs and their hash, the fourth protocol's combined score with the forecast ranges, the fifth protocol's cluster-scale test of the Moster relation with its inputs, the sixth protocol's audit of that test, the seventh protocol's comparison with the published lensing masses, with the transcribed table, the eighth protocol's lensing check of galaxy halos, with its transcribed table, and the ninth protocol's forward model of the KiDS lensing profiles, with the release's files, their hashes and the digitized stellar-mass histograms, and post-run checks that hold out each of the 85 dwarfs in turn. It also records two coding errors in its own checks, found by review after the runs and fixed; neither changes a number reported here. It credits the SPARC, nearby-galaxy, X-COP, CLASH, MaNGA, KiDS and stellar-history catalogs on which it rests. This work collects no new observations.

References

Brouwer, M. M., et al. (2021). The weak lensing radial acceleration relation: Constraining modified gravity and cold dark matter theories with KiDS-1000. *Astronomy & Astrophysics*, 650, A113. https://doi.org/10.1051/0004-6361/202040108

Bruce, J., Li, T. S., Pace, A. B., Heiger, M., Song, Y.-Y., & Simon, J. D. (2023). Spectroscopic analysis of Milky Way outer halo satellites: Aquarius II and Boötes II. *Astrophysical Journal*, 950, 167. https://doi.org/10.3847/1538-4357/acc943

Chae, K.-H., Desmond, H., Lelli, F., McGaugh, S. S., & Schombert, J. M. (2021). Testing the strong equivalence principle. II. Relating the external field effect in galaxy rotation curves to the large-scale structure of the universe. *Astrophysical Journal*, 921, 104. https://doi.org/10.3847/1538-4357/ac1bba

Charlton, P. J. L., Hudson, M. J., Balogh, M. L., & Khatri, S. (2017). The dependence of halo mass on galaxy size at fixed stellar mass using weak lensing. *Monthly Notices of the Royal Astronomical Society*, 472, 2367-2387. https://doi.org/10.1093/mnras/stx2068

Collins, M. L. M., et al. (2013). A kinematic study of the Andromeda dwarf spheroidal system. *Astrophysical Journal*, 768, 172. https://doi.org/10.1088/0004-637X/768/2/172

Cooke, K. C., O'Dea, C. P., Baum, S. A., Tremblay, G. R., Cox, I. G., & Gladders, M. D. (2016). Star formation in intermediate redshift $0.2 < z < 0.7$ brightest cluster galaxies. *Astrophysical Journal*, 833, 224. https://doi.org/10.3847/1538-4357/833/2/224

De Martino, I., Della Monica, R., & Roshan, M. (2025). Dwarf galaxies in non-local gravity. *Physics of the Dark Universe*, 49, 102044. https://doi.org/10.1016/j.dark.2025.102044

Di Cintio, A., Brook, C. B., Dutton, A. A., Macciò, A. V., Stinson, G. S., & Knebe, A. (2014). A mass-dependent density profile for dark matter haloes including the influence of galaxy formation. *Monthly Notices of the Royal Astronomical Society*, 441, 2986. https://doi.org/10.1093/mnras/stu729

Diemer, B. (2018). COLOSSUS: A Python toolkit for cosmology, large-scale structure, and dark matter halos. *Astrophysical Journal Supplement Series*, 239, 35. https://doi.org/10.3847/1538-4365/aaee8c

Dutton, A. A., & Macciò, A. V. (2014). Cold dark matter haloes in the Planck era: Evolution of structural parameters for Einasto and NFW profiles. *Monthly Notices of the Royal Astronomical Society*, 441, 3359. https://doi.org/10.1093/mnras/stu742

Eckert, D., Ettori, S., Pointecouteau, E., van der Burg, R. F. J., & Loubser, S. I. (2022). The gravitational field of X-COP galaxy clusters. *Astronomy & Astrophysics*, 662, A123. https://doi.org/10.1051/0004-6361/202142507

Ettori, S., et al. (2019). Hydrostatic mass profiles in X-COP galaxy clusters. *Astronomy & Astrophysics*, 621, A39. https://doi.org/10.1051/0004-6361/201833323

Famaey, B., Pizzuti, L., & Saltas, I. D. (2025). Nature of the missing mass of galaxy clusters in MOND: The view from gravitational lensing. *Physical Review D*, 111, 123042. https://doi.org/10.1103/dccw-srks

Haghi, H., & Amiri, V. (2016). Testing modified gravity with dwarf spheroidal galaxies. *Monthly Notices of the Royal Astronomical Society*, 463, 1944. https://doi.org/10.1093/mnras/stw2140

Hudson, M. J., et al. (2015). CFHTLenS: Co-evolution of galaxies and their dark matter haloes. *Monthly Notices of the Royal Astronomical Society*, 447, 298-314. https://doi.org/10.1093/mnras/stu2367

Kravtsov, A. V., Vikhlinin, A. A., & Meshcheryakov, A. V. (2018). Stellar mass-halo mass relation and star formation efficiency in high-mass halos. *Astronomy Letters*, 44, 8-34. https://doi.org/10.1134/S1063773717120015

Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves. *Astronomical Journal*, 152, 157. https://doi.org/10.3847/0004-6256/152/6/157

Lelli, F., McGaugh, S. S., Schombert, J. M., & Pawlowski, M. S. (2017). One law to rule them all: The radial acceleration relation of galaxies. *Astrophysical Journal*, 836, 152. https://doi.org/10.3847/1538-4357/836/2/152

Li, P., Lelli, F., McGaugh, S. S., & Schombert, J. M. (2019). A constant characteristic volume density of dark matter haloes from SPARC rotation curve

fits. *Monthly Notices of the Royal Astronomical Society*, 482, 5106. https://doi.org/10.1093/mnras/sty2968

Li, P., Lelli, F., McGaugh, S. S., & Schombert, J. M. (2020). A comprehensive catalog of dark matter halo models for SPARC galaxies. *Astrophysical Journal Supplement Series*, 247, 31. https://doi.org/10.3847/1538-4365/ab700e

Mandelbaum, R., Wang, W., Zu, Y., White, S., Henriques, B., & More, S. (2016). Strong bimodality in the host halo mass of central galaxies from galaxy-galaxy lensing. *Monthly Notices of the Royal Astronomical Society*, 457, 3200-3218. https://doi.org/10.1093/mnras/stw188

McConnachie, A. W. (2012). The observed properties of dwarf galaxies in and around the Local Group. *Astronomical Journal*, 144, 4. https://doi.org/10.1088/0004-6256/144/1/4

McConnachie, A. W., & Côté, P. (2010). Revisiting the influence of unidentified binaries on velocity dispersion measurements in ultra-faint stellar systems. *Astrophysical Journal Letters*, 722, L209. https://doi.org/10.1088/2041-8205/722/2/L209

McGaugh, S., & Milgrom, M. (2013). Andromeda dwarfs in light of modified Newtonian dynamics. *Astrophysical Journal*, 766, 22. https://doi.org/10.1088/0004-637X/766/1/22

McGaugh, S. S., Lelli, F., & Schombert, J. M. (2016). The radial acceleration relation in rotationally supported galaxies. *Physical Review Letters*, 117, 201101. https://doi.org/10.1103/PhysRevLett.117.201101

Minor, Q. E., Pace, A. B., Marshall, J. L., & Strigari, L. E. (2019). Robust velocity dispersion and binary population modelling of the ultrafaint dwarf galaxy Reticulum II. *Monthly Notices of the Royal Astronomical Society*, 487, 2961-2968. https://doi.org/10.1093/mnras/stz1468

Moffat, J. W., & Rahvar, S. (2013). The MOG weak field approximation and observational test of galaxy rotation curves. *Monthly Notices of the Royal Astronomical Society*, 436, 1439. https://doi.org/10.1093/mnras/stt1670

Moliné, Á., Sánchez-Conde, M. A., Palomares-Ruiz, S., & Prada, F. (2017). Characterization of subhalo structural properties and implications for dark matter annihilation signals. *Monthly Notices of the Royal Astronomical Society*, 466, 4974-4990. https://doi.org/10.1093/mnras/stx026

Moster, B. P., Naab, T., & White, S. D. M. (2013). Galactic star formation and accretion histories from matching galaxies to dark matter haloes. *Monthly Notices of the Royal Astronomical Society*, 428, 3121. https://doi.org/10.1093/mnras/sts261

Navarro, J. F., et al. (2017). The origin of the mass discrepancy-acceleration relation in ΛCDM. *Monthly Notices of the Royal Astronomical Society*, 471, 1841. https://doi.org/10.1093/mnras/stx1705

Navarro, J. F., Frenk, C. S., & White, S. D. M. (1997). A universal density profile from hierarchical clustering. *Astrophysical Journal*, 490, 493-508. https://doi.org/10.1086/304888

Pace, A. B. (2025). The Local Volume Database: a library of the observed properties of nearby dwarf galaxies and star clusters. *The Open Journal of Astrophysics*, 8. https://doi.org/10.33232/001c.144859

Pace, A. B., Erkal, D., & Li, T. S. (2022). Proper motions, orbits, and tidal influences of Milky Way dwarf spheroidal galaxies. *Astrophysical Journal*, 940, 136. https://doi.org/10.3847/1538-4357/ac997b

Pantos, I., & Perivolaropoulos, L. (2026). Status of the S8 tension: A 2026 review of probe discrepancies. arXiv:2602.12238.

Rahvar, S., & Mashhoon, B. (2014). Observational tests of nonlocal gravity: Galaxy rotation curves and clusters of galaxies. *Physical Review D*, 89, 104011. https://doi.org/10.1103/PhysRevD.89.104011

Read, J. I., Agertz, O., & Collins, M. L. M. (2016). Dark matter cores all the way down. *Monthly Notices of the Royal Astronomical Society*, 459, 2573. https://doi.org/10.1093/mnras/stw713

Simon, J. D., et al. (2020). Birds of a feather? Magellan/IMACS spectroscopy of the ultra-faint satellites Grus II, Tucana IV, and Tucana V. *Astrophysical Journal*, 892, 137. https://doi.org/10.3847/1538-4357/ab7ccb

Skordis, C., & Zlosnik, T. (2021). New relativistic theory for modified Newtonian dynamics. *Physical Review Letters*, 127, 161302. https://doi.org/10.1103/PhysRevLett.127.161302

Taylor, E. N., et al. (2020). GAMA + KiDS: Empirical correlations between halo mass and other galaxy properties near the knee of the stellar-to-halo mass relation. *Monthly Notices of the Royal Astronomical Society*, 499, 2896-2911. https://doi.org/10.1093/mnras/staa2648

Tian, Y., Ko, C.-M., Li, P., McGaugh, S., & Poblete, S. L. (2024). A distinct radial acceleration relation across brightest cluster galaxies and galaxy clusters. *Astronomy & Astrophysics*, 684, A180. https://doi.org/10.1051/0004-6361/202347868

Tian, Y., Umetsu, K., Ko, C.-M., Donahue, M., & Chiu, I.-N. (2020). The radial acceleration relation in CLASH galaxy clusters. *Astrophysical Journal*, 896, 70. https://doi.org/10.3847/1538-4357/ab8e3d

Tinker, J., et al. (2008). Toward a halo mass function for precision cosmology: The limits of universality. *Astrophysical Journal*, 688, 709-728. https://doi.org/10.1086/591439

Umetsu, K., Zitrin, A., Gruen, D., Merten, J., Donahue, M., & Postman, M. (2016). CLASH: Joint analysis of strong-lensing, weak-lensing shear, and magnification data for 20 galaxy clusters. *Astrophysical Journal*, 821, 116. https://doi.org/10.3847/0004-637X/821/2/116

van der Wel, A., et al. (2014). 3D-HST+CANDELS: The evolution of the galaxy size-mass distribution since z = 3. *Astrophysical Journal*, 788, 28. https://doi.org/10.1088/0004-637X/788/1/28

Wolf, J., et al. (2010). Accurate masses for dispersion-supported galaxies. *Monthly Notices of the Royal Astronomical Society*, 406, 1220-1237. https://doi.org/10.1111/j.1365-2966.2010.16753.x